

Shivashankar S Menon

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WORK EXPERIENCE

Founding Engineer - AI @ OnFinance AI

Sep 2023 - Mar 2024

- Built the **retrieval/RAG** side end-to-end — ingestion, indexing, retrieval, reranking, generation — and drove the applied research on retrieval quality.
- Shipped RAG document-QA pipelines for **10+ financial-sector clients** (Kotak, ICICI, IBSFintech, Nuvama/Edelweiss, Oister, NSE); the **LetsVenture** engagement converted to a contracted production deployment.
- Lifted top-1 retrieval recall **25% → 90%** on an internal FAQ set via question-embedding indexing; cut query latency **up to 75%** with chunk-budget tuning and a cost-aware GPT-4-Turbo/3.5 switch.
- Built **two-stage retrieval** (broad first pass, then a more accurate re-rank), **GPT-4V + Azure Document Intelligence** ingestion to recover tables/charts OCR missed, and short/long-term chat memory for conversational querying.
- Led **retrieval-quality research** — entity-attribute question generation, atomicity-based chunking, and Bloom-tier answer routing — later distilled into a public writeup series.

ML Engineer Intern @ American Express

Jul 2022 - Dec 2022

Member of R&D team in-charge of developing Distributed GPU Learning modes for AmEx proprietary XGBoost algorithms. Tested the new modes throughout the development cycle on internal datasets and analyzed them on GINI, AUC, train/turnaround times, Capture Rate at 20 percentile etc — New modes trained models 20x faster and readied them 5x faster than the then-most widely-used mode. Shipped as an official feature update (v3.0, Nov 2022) in the company's algorithm.

R&D Intern @ IIRS Dehradun

Jun 2021 - Jul 2021

Implemented a satellite image segmentation model based on the U-Net Architecture using Keras. Analyzed the structure of the Michigan Microwave Canopy Scattering ML Models for simulating microwave back-scattering across forest canopies.

INDEPENDENT PURSUITS

Civil Services (UPSC) Preparation

2024 - 2026

- Self-directed preparation for India's Civil Services (UPSC) examination. I built a holistic understanding of current and historical socio-economic issues, the digital divide, and the scope for AI across Indian industries. Extensive research into public policy and governance sharpened how I analyse complex, real-world problems.
- **Research & Analysis** – synthesised large volumes of information across Economics, Public Administration, and International Relations.
- **Policy & Writing** – broke down complex real-world issues and drafted analytical essays and reports under time constraints.
- **Discipline & Rigor** – sustained daily study routines reinforcing focus, self-management, and high-pressure endurance.

PROJECTS

Multi-Agent LLM Simulation – Amphoreus

[Demo Link](#)

Designed and built a fully automated multi-agent LLM system (Dungeon Master + **Orchestrator** + players) that runs a D&D campaign via a structured engine loop — **tool-only state mutation**, typed vector-indexed memory, **provider independence** — plus a variance-aware harness for evaluating non-deterministic agent behavior.

Accessible Wheelchair Routing

[Publication Link](#)

Created **two datasets of vibration data** gathered from manual/powered wheelchairs on fifteen surface types. Implemented a surface classification model on the data using **adaptive activation functions**; extended said framework to the powered wheelchair (PW) dataset via **transfer learning** with 98.8% accuracy. **Models outperformed other SOTA models by 3% with 40% lesser train-time.**

Foreground Emissions Modeling for Astronomy

[Publication Link](#)

Parsed, analyzed Deep-Sky Observation data gathered from Galaxy Evolution Explorer (GALEX) satellite’s spacecraft state files. Derived empirical model for foreground emissions involving **Robust Regression** concepts. Generalized the model to Deep and Medium-Sky Observations.

EDUCATION

2019 - 2023	B.E. CS (with DS Minor) at BITS Pilani, Goa Campus	(CGPA: 9.15/10)
2018 - 2019	Class 12th CBSE	(92.8%)
2016 - 2017	Class 10th CBSE	(CGPA: 10/10)

PUBLICATIONS

- N. Narayanan, S.S. Menon et. al. (Jan. 1, 2023a). “A model of foreground emission in UV using GALEX deep observations”. In: *Advances in Space Research* 71.1, pp. 1059–1073. ISSN: 0273-1177. DOI: [10.1016/j.asr.2022.07.086](#). URL: <https://www.sciencedirect.com/science/article/pii/S0273117722007542>.
- (May 30, 2023b). “Validation of Foreground Emission Model using GALEX Medium Observations”. In: *Research in Astronomy and Astrophysics* 23.6. Publisher: National Astronomical Observatories, CAS and IOP Publishing, p. 065021. ISSN: 1674-4527. DOI: [10.1088/1674-4527/acd09f](#). URL: <https://doi.org/10.1088/1674-4527/acd09f>.
- V. Misra, S.S. Menon et. al. (Aug. 24, 2024). “AdaGen: Adaptive Generalized Knowledge Transfer Framework for Sensor-Based Surface Classification for Wheelchair Routing”. In: *SN Computer Science* 5.7, p. 820. ISSN: 2661-8907. DOI: [10.1007/s42979-024-03181-w](#). URL: <https://doi.org/10.1007/s42979-024-03181-w>.

SKILLS

ML/DL/AI	Numpy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Claude Code
Programming	OOP, DBMS, DSA
Languages	Python, $\text{\LaTeX}$
Policy & Analysis	Public Policy, Economics, Public Administration, International Relations, Good Governance, Research & Synthesis, Analytical Writing